Filters

Intro to Audio Tech 1

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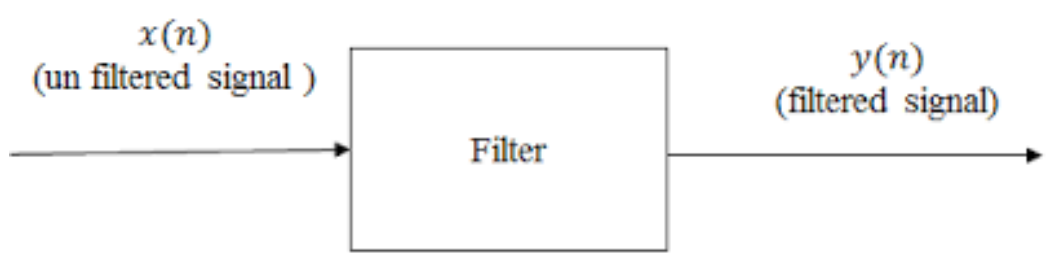
Lesson Outline

- Filters
  - Types
  - Design
  - Characterizing
- Max Demo
- Up Next…

Filters

What is a filter?

- A function that takes in a signal and does something to it
- Often it will output something “less” than before
- A filter selectively attenuates specific frequency components of a signal
  - (It also modifies the relative phase of components)
- When working with filters, we care a lot about how frequencies change between our input and output signals



Common Types of Filters

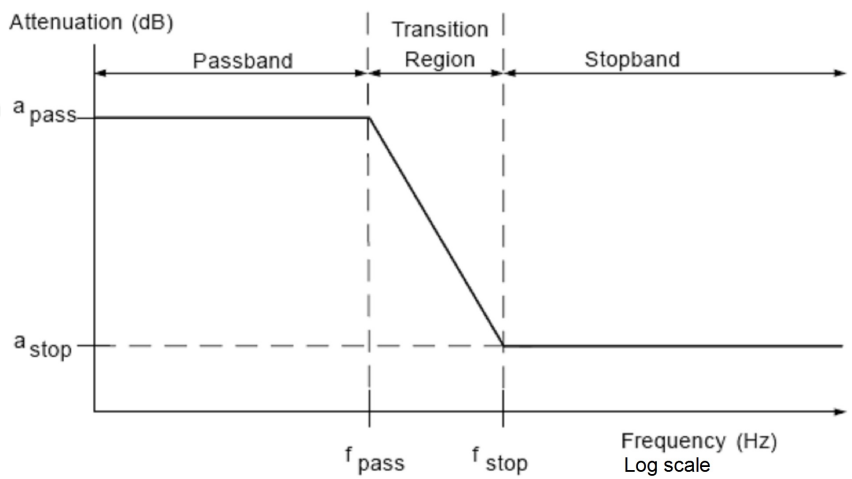
- Low-pass
- High-pass
- Band-reject
- Band-pass
- Other filter types
  - Low-shelf, high-shelf, comb, all-pass
  - Specific topologies - Bessel, Butterworth, Chebyshev, elliptical filters

Filter Terms

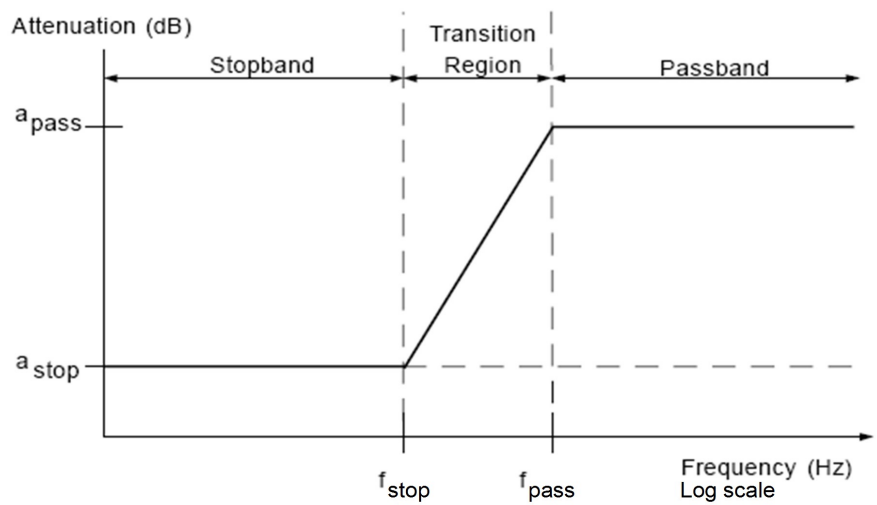
- Frequency Response
  - How the signal changes over every frequency
    - Magnitude, phase response
- Cutoff Frequency (half power point)
  - The frequency point where the signal is attenuated to half the power
- Passband and Stopband
  - Pass - range of frequencies that pass through the filter (are unaffected)
  - Stop - range of frequencies that are stopped by the filter (are filtered)
- Transition Band
  - Range of frequencies between the pass/stopbands
  - Characterized by the center frequency, bandwidth/resonance/Q, and slope

Low-Pass and High-Pass

- Low pass allows low frequencies through under the cutoff frequency

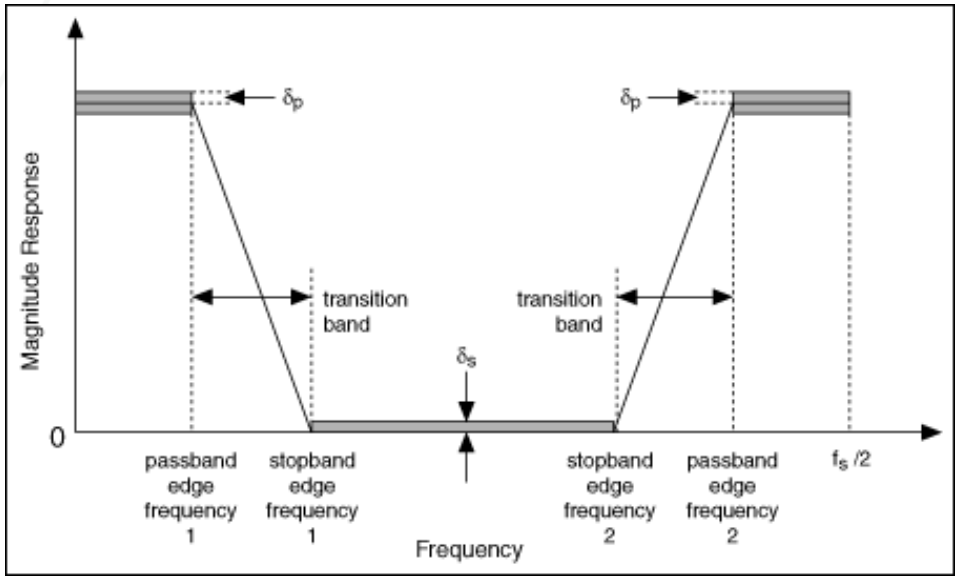


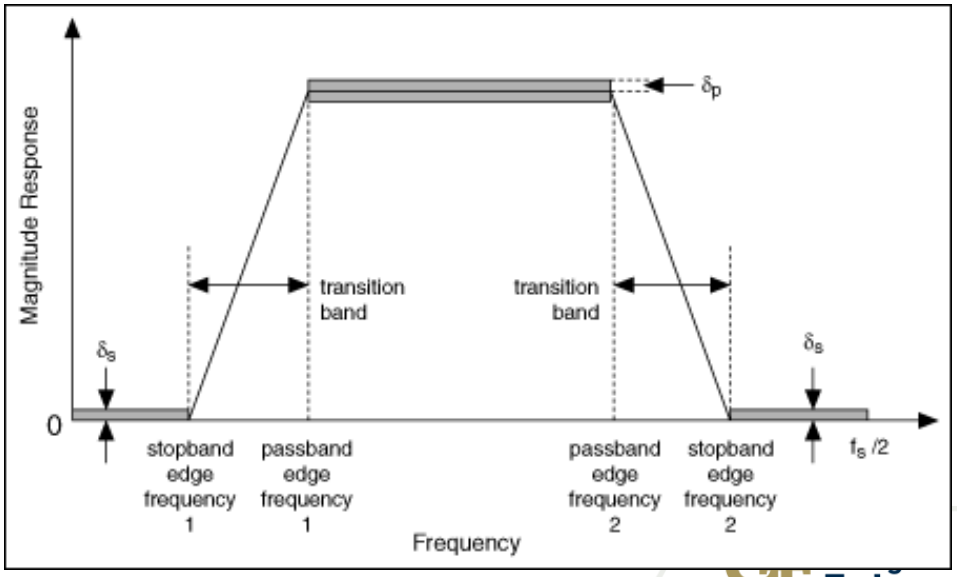
- High pass allows high frequencies through above the cutoff frequency



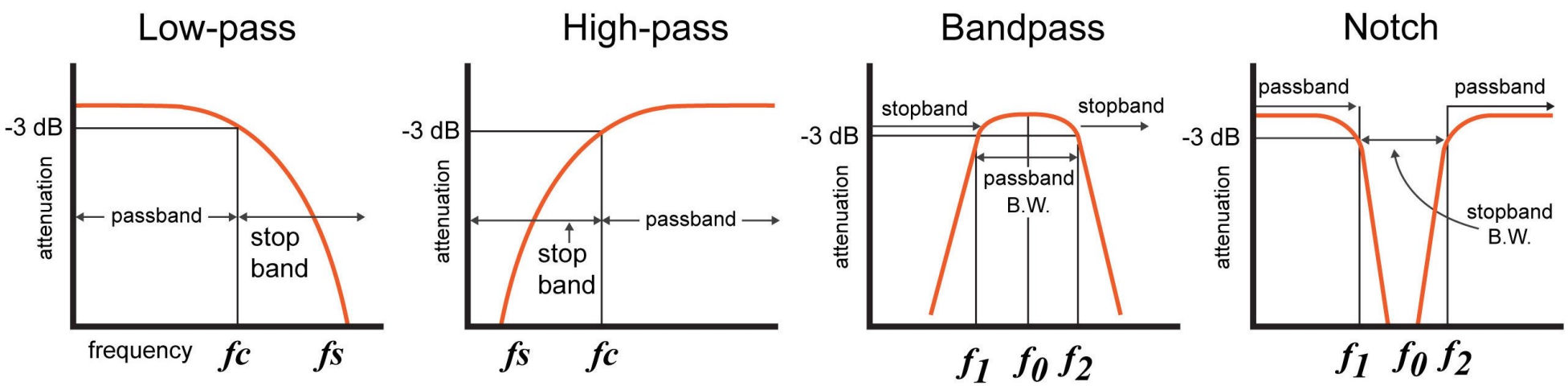
Band-Reject and Band-Pass

- Band reject stops between range of frequencies
  - Can create a notch with narrow band
- Band pass allows frequencies above and below a certain point
  - Can create a peak with narrow band





Four Basic Filters



What is Q?

- Q factor or quality factor sets the resonance of the filter
- You can define a filter's response by the cutoff frequencies, bandwidth and center frequency, or by the center frequency and Q
- Q = center frequency / bandwidth
  - Q and bandwidth are inversely proportional
- Q factor is often used in EQ and other filter plugins in the "real world"

Why use Q?

- We use Q to help our perceptual understanding of a filter
  - A passband of 300 Hz with a center frequency of 200 Hz (cutoff frequencies at 100 Hz and 400 Hz) will sound very wide (2 octaves apart)
  - However, the same passband of 300 Hz with a center frequency of 5198 Hz (cutoff frequencies at 5050 Hz and 5350 Hz) will sound narrow (one semitone apart)
  - The Q factor is more consistent with the perception
    - Q = 200 / 300 = 0.66
    - Q = 5198 / 300 = 17.3
- Higher Q -> narrower filter
- Lower Q -> wider filter

Filter Design

Aside: Properties of LTI Systems

- LTI – Linear Time-Invariant
- Time Invariance
  - Output doesn't depend on when the input is sent
  - No matter the time of input, the system input and output relationship will be the same
- Linearity
  - The sum of inputs is the sum of the outputs of individual inputs
  - Scalar gains do not affect the output
  - $y[n] = A \cdot x_a[n] + B \cdot x_b[n] \leftrightarrow A \cdot y_a[n] + B \cdot y_b[n]$
- Causality
  - The output is only dependent on the past or current input
  - Nothing in the future impacts the system
- Consideration for filters:
  - Stability
    - System is stable if its output is bounded
    - Can't "explode" with "feedback"

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Simplest Possible Filter

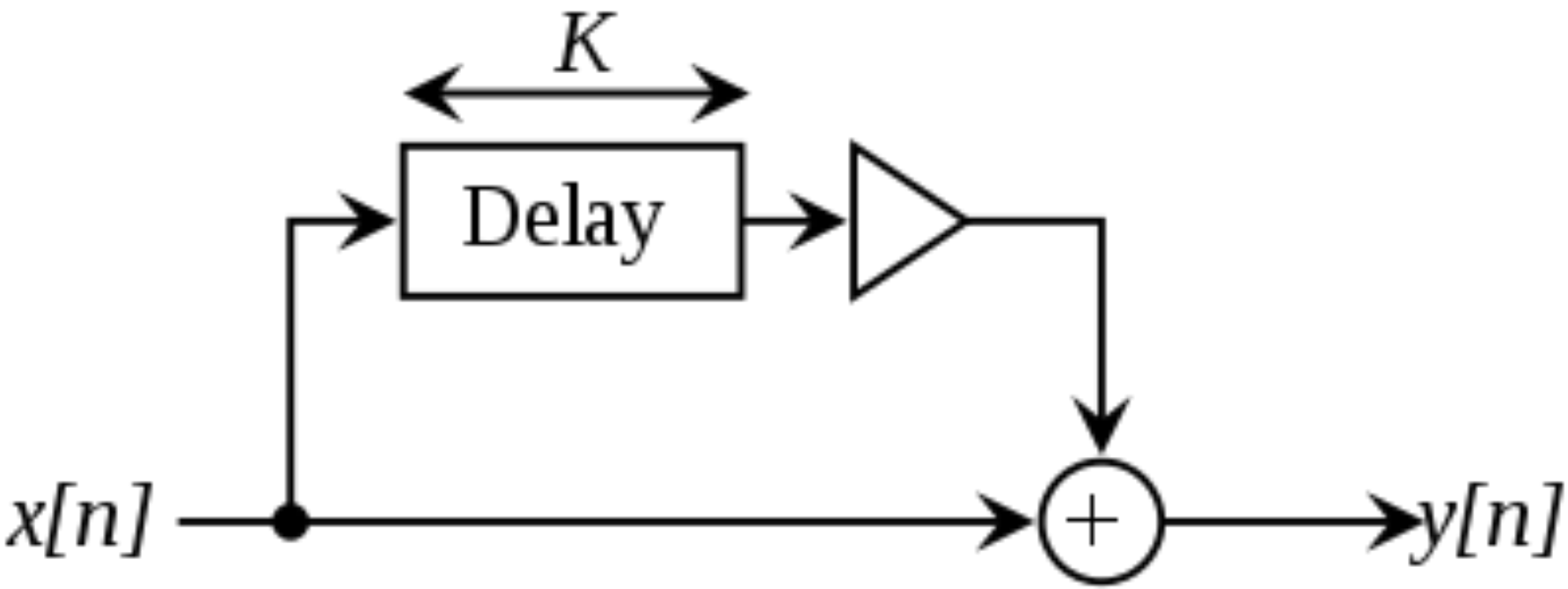
- Scaled signal x[n] added to some scaled delayed version of itself

$$y_t = a_0 x_t + a_1 x_{t-\tau}$$

$$y[n] = a_0 \mathrm{x}[\mathrm{n}] + a_1 x[n-1]$$

- We ”plug-in” numbers sample by sample

Simplest Possible Filter – Block Diagram



Filter Order

- We are delaying and scaling by samples at a time
- If a filter has a one-sample delay, it's called a first-order filter

$$y[n] = a_0 \mathrm{x}[\mathrm{n}] + a_1 x[n-1]$$

- If a filter has a two-sample delay, it's called a second-order filter

$$y[n] = a_0 \mathrm{x}[\mathrm{n}] + a_1 x[n-1] + a_2 x[n-2]$$

What does that mean practically?

- A perfect first order filter attenuates by 6dB every octave
- For example, a first-order low pass filter with a cutoff frequency of 300 Hz yields

Amplitude Change	Frequency	Frequency	Frequency	Frequency
0 dB	< 300 Hz			
-6 dB		600 Hz		
-12 dB			1200 Hz	
-18 dB				2400 Hz

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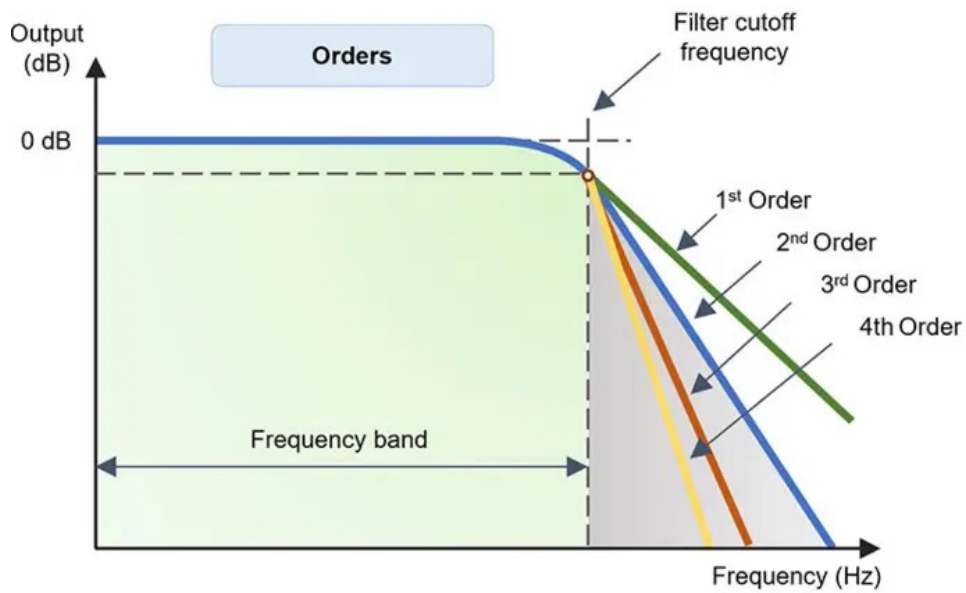
What does that mean practically?

- A perfect second-order filter attenuates by 12 dB every octave
- For example, a second-order low pass filter with a cutoff frequency of 300 Hz yields

Amplitude Change	Frequency	Frequency	Frequency	Frequency
0 dB	< 300 Hz			
-6 dB				
-12 dB		600 Hz		
-18 dB				
-24 dB			1200 Hz	
-30 dB				
-36 dB				2400 Hz

What does that mean practically?

- The higher the order, the steeper the transition from the pass and stopband
- The implementation of digital filters is not perfect, so there is a transition zone
- Meaning though the cutoff shouldn't be impacted (0 dB change), usually there is about a -3 dB change at the cutoff frequency
- The octave attenuation is a good reference, but for filters of 2+ order, it will not always hold up



So, what defines the cutoff and filter type?

- The coefficients in the difference equation determine the behavior of the filter
- These are tricky to calculate (and we will not do it by hand)
- It is important to note, that the cutoff frequency depends on the sampling rate along with the coefficients

- Low pass filter with cutoff at ¼ the sampling rate

$$y[n] = \frac{1}{2}\mathrm{x[n]} + \frac{1}{2}x[n-1] \text{ (this is also a moving average)}$$

- High pass filter with cutoff of ¼ the sampling rate

$$y[n] = \frac{1}{2}\mathrm{x[n]} - \frac{1}{2}x[n-1]$$

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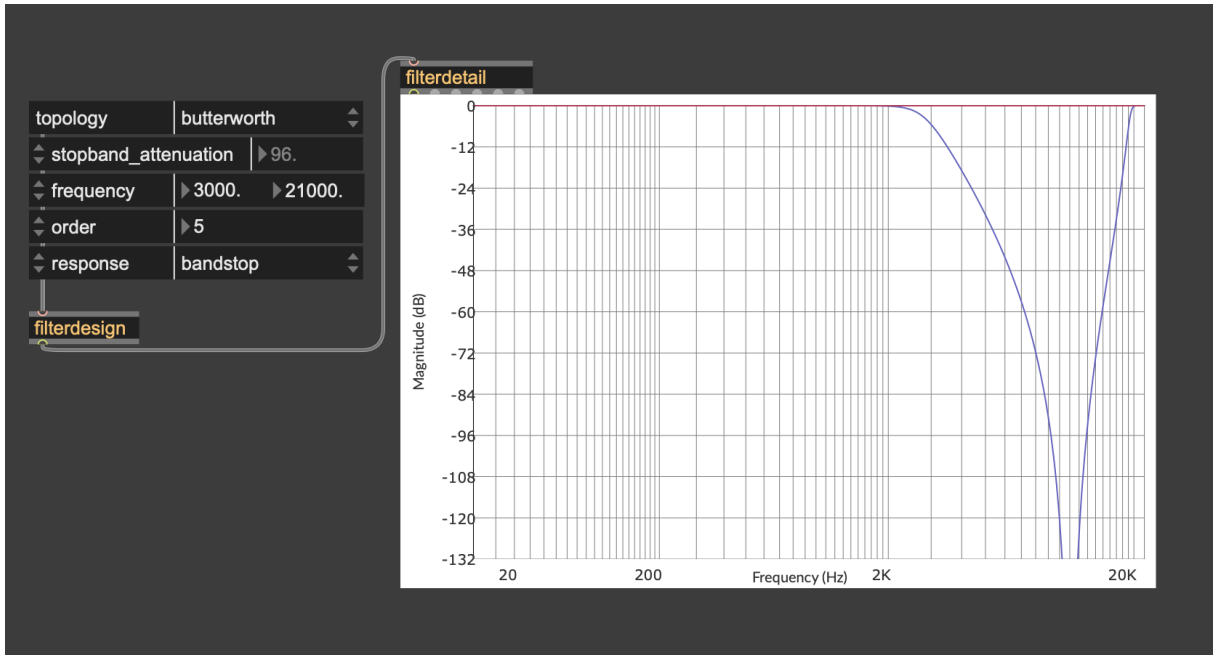
Understanding Coefficients and Filters

- The filterdesign object creates a filter based on filter specifications like cutoff, order, and response type



Characterizing Filters

Impulse Response

- We can send an impulse through our filter to determine how it reacts in the **time domain**
- A perfect impulse is a short burst signal containing all frequencies within a sampling rate
  - This looks like a signal of [1, 0, 0, 0, 0, 0…]
- For LTIs, the impulse response fully characterizes the filter
- For the "Simplest Possible Filter":

$$y[n] = \mathrm{x[n]} + x[n-1]$$
$$\mathrm{h}[n] = \{1, 1, 0, 0, 0, 0..\}$$

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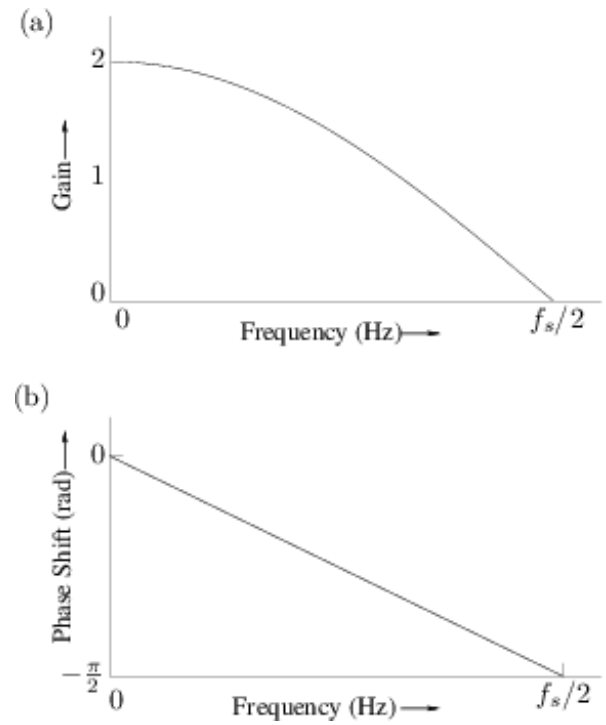
Frequency Response

- We can send the same unit impulse through a filter to determine how it acts in the **frequency domain**
- However, this time instead of looking at how the input changes over time, we are looking at how the spectrum changes
- The unit impulse contains a flat spectrum of all frequencies, so we find the frequency response to determine how those frequencies are attenuated
- You'll see this in two graphs: magnitude frequency response (a) and phase response (b)



Impulse vs Frequency Response

- Impulse Response
  - The filter's response to a unit impulse – how it characterizes that impulse
  - "How are the outputted samples affected"
  - Time domain
- Frequency Response
  - The filter's response to a unit impulse
  - How does the system respond to specific frequencies
  - Frequency domain

Finite Impulse Response vs Infinite Impulse Response

- FIR
  - Simple delay line
  - Filter's response to an impulse dies over a **finite** period of time
  - Pros: "Easy" design, stable, linear phase
  - Cons: Not efficient, more memory used and longer delay, requires high order for "good" filtering
  - Example FIR High pass Filter:
    - $y[n] = x[n] - x[n-1]$
- IIR
  - Feedback, recursive
  - Filter's response feeds back on itself
  - Pros: efficient, lower order for sharp filtering, quick response
  - Cons: Unstable (can "blow up"), phase distortion and nonlinear phase
  - Example IIR High pass Filter:
    - $y[n] = x[n] - y[n-1]$

Filters in Max